Questions

Q1.

(a) Using the substitution $u = \sqrt{2x+1}$, show that

$$\int_4^{12} \sqrt{8x+4}\, e^{\sqrt{2x+1}}\, dx$$

may be expressed in the form

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$$\int_a^b ku^2 e^u\, du$$

where a, b and k are constants to be found.

(4)

(b) Hence find, by algebraic integration, the exact value of

$$\int_4^{12} \sqrt{8x+4}\, e^{\sqrt{2x+1}}\, dx$$

giving your answer in simplest form.

(5)

(Total for question = 9 marks)

(Q04 WMA14/01, Jan 2023)

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Q2.

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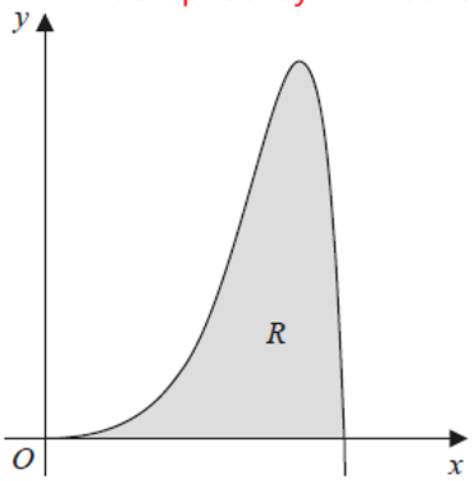


Figure 2

(a) Find $\int e^{2x} \sin x \, dx$

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(5)

Figure 2 shows a sketch of part of the curve with equation

$$y = e^{2x} \sin x \quad x \geq 0$$

The finite region R is bounded by the curve and the x-axis and is shown shaded in Figure 2.

(b) Show that the exact area of R is $\frac{e^{2\pi} + 1}{5}$

(Solutions relying on calculator technology are not acceptable.)

(2)

(Total for question = 7 marks)

(Q07 WMA14/01, Jan 2021)

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Q3.

In this question you must show all stages of your working.

Solutions based on calculator technology are not acceptable.

(i) Use integration by parts to find the exact value of

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$$\int_0^4 x^2 e^{2x}\,dx$$

giving your answer in simplest form.

(5)

(ii) Use integration by substitution to show that

$$\int_3^{\frac{21}{2}} \frac{4x}{(2x-1)^2}\,dx = a + \ln b$$

where *a* and *b* are constants to be found.

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(7)

(Total for question = 12 marks)

(Q03 WMA14/01, Oct 2023)

Q4.

(i) Find

$$\int x^2 e^x dx$$

(4)

(ii) Use the substitution $u = \sqrt{1-3x}$ to show that

$$\int \frac{27x}{\sqrt{1-3x}}dx = -2(1-3x)^{\frac{1}{2}}(Ax+B)+k$$

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where *A* and *B* are integers to be found and *k* is an arbitrary constant.

(6)

(Total for question = 10 marks)

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(Q05 WMA14/01, June 2023)

Q5.

In this question you should show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Using the substitution $u = 3 + \sqrt{2x - 1}$ find the exact value of

$$\int_1^{13} \frac{4}{3 + \sqrt{2x - 1}} \, dx$$

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giving your answer in the form $p + q \ln 2$, where p and q are integers to be found.

(8)

(Total for question = 8 marks)

(Q05 WMA14/01, Jan 2021)

Q6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

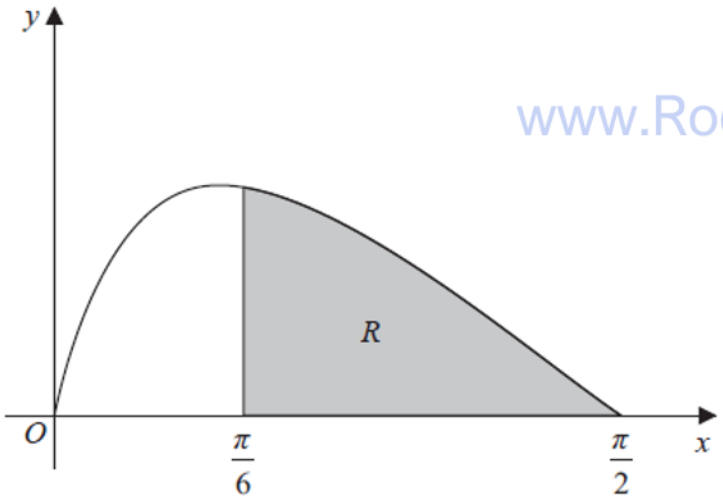


Figure 2

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Figure 2 shows a sketch of the curve with equation

$$y = \frac{16\sin 2x}{(3+4\sin x)^2} \qquad 0 \leqslant x \leqslant \frac{\pi}{2}$$

The region R, shown shaded in Figure 2, is bounded by the curve, the x-axis and the line with equation $x = \frac{\pi}{6}$

Using the substitution $u = 3 + 4\sin x$, show that the area of R can be written in the form $a + \ln b$, where a and b are rational constants to be found.

(7)

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(Total for question = 7 marks)

(Q06 WMA14/01, Oct 2021)

Q7.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Use the substitution $x = 2\sin u$ to show that

$$\int_0^1 \frac{3x+2}{(4-x^2)^{\frac{3}{2}}}\,\mathrm{d}x = \int_0^p \left(\frac{3}{2}\sec u\tan u + \frac{1}{2}\sec^2 u\right)\mathrm{d}u$$

where p is a constant to be found.

(b) Hence find the exact value of

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$$\int_0^1 \frac{3x+2}{(4-x^2)^{\frac{3}{2}}}\,\mathrm{d}x$$

(4)

(Total for question = 4 marks)

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(Q05 WMA14/01, June 2022)

Q8.

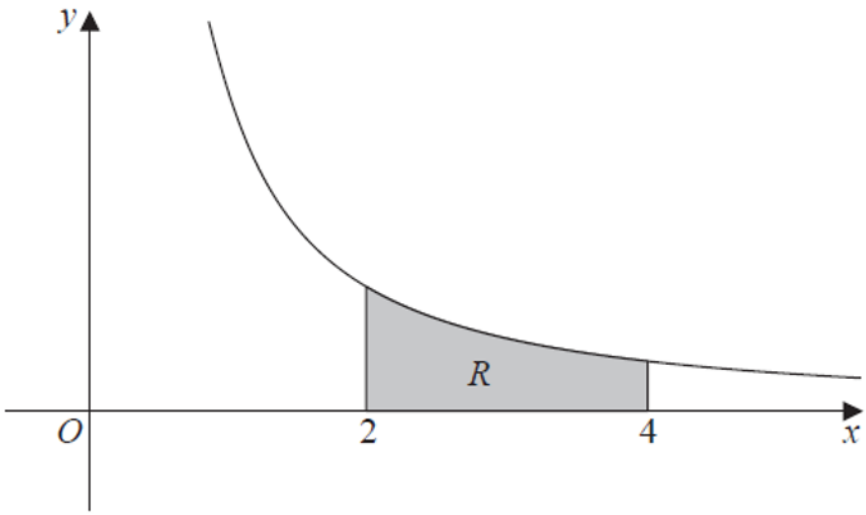


Figure 3

(a) Find $\int \frac{\ln x}{x^2} dx$

(3)

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{3 + 2x + \ln x}{x^2} \qquad x > 0.5$$

The finite region R, shown shaded in Figure 3, is bounded by the curve, the line with equation $x = 2$, the x-axis and the line with equation $x = 4$

(b) Use the answer to part (a) to find the exact area of R, writing your answer in simplest form.

(4)

(Total for question = 7 marks)

(Q05 WMA14/01, Oct 2020)

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Q9.

(i) Using a suitable substitution, find, using calculus, the value of

$$\int_1^5 \frac{3x}{\sqrt{2x-1}}\,\mathrm{d}x$$

(*Solutions relying entirely on calculator technology are not acceptable.*)

(6)

(ii) Find

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$$\int \frac{6x^2-16}{(x+1)(2x-3)}\,\mathrm{d}x$$

(6)

(Total for question = 12 marks)

(Q07 WMA14/01, Oct 2020)

Q10.

Use algebraic integration and the substitution $u = \sqrt{x}$ to find the exact value of

$$\int_1^4 \frac{10}{5x+2x\sqrt{x}}\,\mathrm{d}x$$

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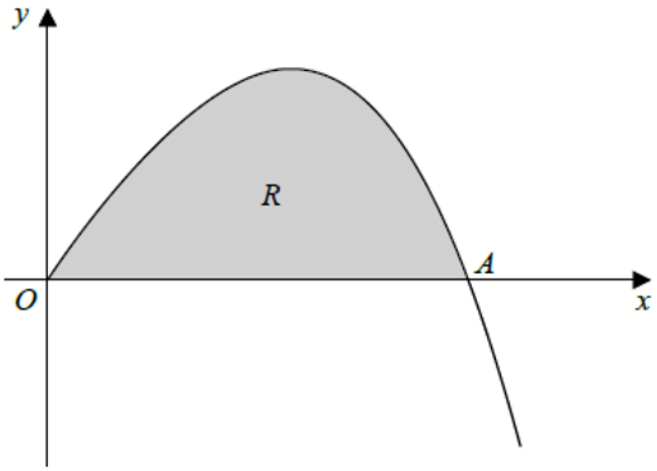
Write your answer in the form $4\ln\left(\frac{a}{b}\right)$, where a and b are integers to be found.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(Total for question = 8 marks)

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Q11.



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Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = 4x - xe^{\frac{1}{2}x}, x \geqslant 0$

The curve meets the x-axis at the origin O and cuts the x-axis at the point A.

(a) Find, in terms of ln 2, the x coordinate of the point A.

(2)

(b) Find $\int xe^{\frac{1}{2}x} dx$

(3)

The finite region R, shown shaded in Figure 2, is bounded by the x-axis and the curve with equation

$y = 4x - xe^{\frac{1}{2}x}, x \geqslant 0$

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(c) Find, by integration, the exact value for the area of R.

Give your answer in terms of ln 2

(3)

(Total for question = 8 marks)

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(Q05 WMA14/01, Specimen papers)

Q12.

(a) Find the derivative with respect to y of

$$\frac{1}{(1+2\ln y)^2}$$

(2)

(b) Hence find a general solution to the differential equation

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$$3\text{cosec}(2x)\frac{dy}{dx} = y(1+2\ln y)^3 \qquad y>0 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

(4)

(c) Show that the particular solution of this differential equation for which $y = 1$ at $x = \frac{\pi}{6}$ is given by

$$y = e^{A\sec x - \frac{1}{2}}$$

where A is an irrational number to be found.

(5)

(Total for question = 11 marks)

(Q09 WMA14/01, Jan 2022)

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Q13.

$$f(x) = \frac{8x-5}{(2x-1)(4x-3)} \qquad x>1$$

(a) Express f(x) in partial fractions.

(3)

(b) Hence find $\int f(x)\,dx$

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(3)

(c) Use the answer to part (b) to find the value of k for which

$$\int_{k}^{3k} f(x)\,dx = \frac{1}{2}\ln 20$$

(5)

(Total for question = 11 marks)

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(Q03 WMA14/01, June 2023)

Q14.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Use the substitution $u = e^x - 3$ to show that

$$\int_{\ln 5}^{\ln 7} \frac{4e^{3x}}{e^x - 3}\,dx = a + b\ln 2$$

where a and b are constants to be found.

(7)

(ii) Show, by integration, that

$$\int 3e^x \cos 2x\,dx = pe^x \sin 2x + qe^x \cos 2x + c$$

where p and q are constants to be found and c is an arbitrary constant.

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(5)

(Total for question = 12 marks)

(Q07 WMA14/01, Oct 2022)

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Q15.

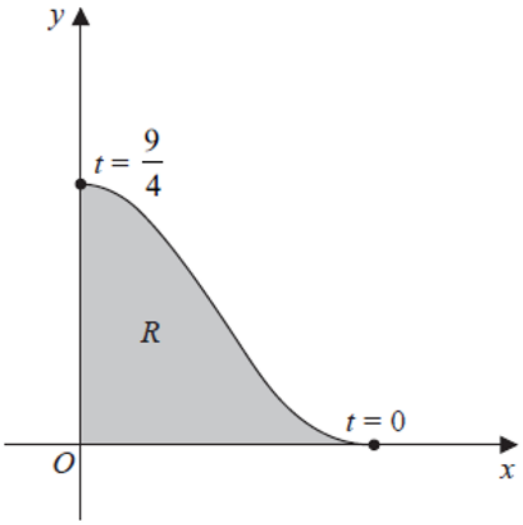


Figure 2

Figure 2 shows a sketch of the curve with parametric equations

$$x = \sqrt{9 - 4t} \qquad y = \frac{t^3}{\sqrt{9 + 4t}} \qquad 0 \leqslant t \leqslant \frac{9}{4}$$

The curve touches the x-axis when $t = 0$ and meets the y-axis when $t = \frac{9}{4}$

The region R, shown shaded in Figure 2, is bounded by the curve, the x-axis and the y-axis.

(a) Show that the area of R is given by

$$K\int_0^{\frac{9}{4}} \frac{t^3}{\sqrt{81 - 16t^2}}\,\mathrm{d}t$$

where K is a constant to be found.

(4)

(b) Using the substitution $u = 81 - 16t^2$, or otherwise, find the exact area of R.

(*Solutions relying on calculator technology are not acceptable.*)

(6)

(Total for question = 10 marks)

(**Q05 WMA14/01, Jan 2022**)

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Q16.

(a) Express $\frac{1}{(1+3x)(1-x)}$ in partial fractions.

(3)

(b) Hence find the solution of the differential equation

$$(1 + 3x)(1 - x)\frac{dy}{dx} = \tan y \qquad -\frac{1}{3} < x \leq \frac{1}{2}$$

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for which $x = \frac{1}{2}$ when $y = \frac{\pi}{2}$

Give your answer in the form $\sin^n y = f(x)$ where n is an integer to be found.

(6)

(Total for question = 9 marks)

(**Q02 WMA14/01, June 2022**)

Q17.

$$f(x) = \frac{1}{x(3x-1)^2} = \frac{A}{x} + \frac{B}{(3x-1)} + \frac{C}{(3x-1)^2}$$

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(a) Find the values of the constants A, B and C

(4)

(b) (i) Hence find $\int f(x)\,dx$

(ii) Find $\int_1^2 f(x)\,dx$, giving your answer in the form $a + \ln b$, where a and b are constants.

(6)

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(Total for question = 10 marks)

(Q03 WMA14/01, Specimen papers)

Q18.

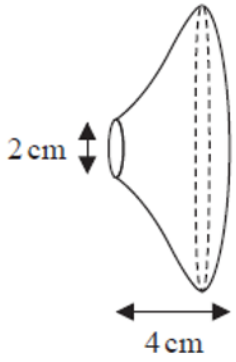


Figure 3

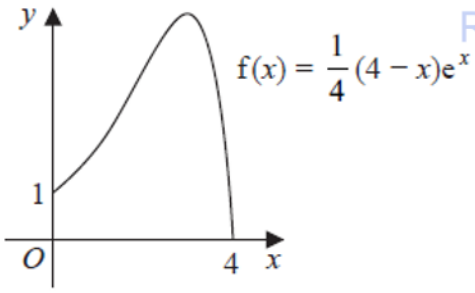


Figure 4

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Figure 3 shows the design of a doorknob.

The shape of the doorknob is formed by rotating the curve shown in Figure 4 through 360° about the x-axis, where the units are centimetres.

The equation of the curve is given by

$$f(x) = \frac{1}{4}(4-x)e^x \qquad 0 \leqslant x \leqslant 4$$

(a) Show that the volume, V cm^3, of the doorknob is given by

$$V = K\int_0^4 (x^2 - 8x + 16)e^{2x}\,dx$$

where K is a constant to be found.

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(3)

(b) Hence, find the exact value of the volume of the doorknob.

Give your answer in the form $p\pi(e^q + r)$cm^3 where p, q and r are simplified rational numbers to be found.

(5)

(Total for question = 8 marks)

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(**Q07 WMA14/01, Jan 2022**)

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